Arham Malik *Software Engineer*

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PROFILE

Software Engineer with 2+ years of experience building scalable Al platforms, web apps, and game systems. Expert in Python (FastAPI), .NET, and React/Next.js, with specialized depth in LangChain and ETL optimization. Proven track record in modernizing legacy architectures into modular, API-driven services that bridge complex data engineering with seamless user experiences.

EDUCATION

Bachelor of Science in Computer Science
BS in Computer Science, FAST NUCES
08/2021 – 06/2025 | Islamabad, Pakistan

Graduated with a Bachelor of Science in Computer Science, specializing in software development, algorithms, and machine learning.

SKILLS

C/C++

Python
Django | Flask | PyTorch | TensorFlow | LangChain & LangSmith | Polars | Pandas | Numpy | Scipy | StreamLit

C#
ASP.NET

Java

JavaScript
React | Node.js

Cloud (GCP)
Cloud Functions | Cloud Run | Pub/Sub | Cloud Scheduler | BigQuery

Database
PostgreSQL | Microsoft SQL Server | MySQL | SQLite | MongoDB | Firebase

PROFESSIONAL EXPERIENCE

Software Engineer
RedCoast Corporation
05/2025 – 11/2025 | Islamabad, Pakistan

- Reduced ETL runtime **15%** by refactoring Python/ BigQuery pipelines and adding partitioning and clustering.
- Cut query latency **60%** and lowered scanned bytes **30%** via query rewrites and cached-results strategy.
- Automated **7+** daily jobs with Cloud Functions/Run, Scheduler, and Pub/Sub.
- Automated **60-day** backfills, improving data freshness/ throughput **60%**.

Junior Software Engineer
RedCoast Corporation
08/2024 – 05/2025 | Islamabad, Pakistan

- Built **8** FastAPI endpoints with **p95 latency ≤ 150 ms** (excluding model inference) for chat and memory workflows.
- Shipped real-time dashboards used by **50** users; reduced analysis time by **50%**.
- Implemented LangChain tools/chains; automated **80%** of routine prompts and reduced support requests by **30%**.

Software Trainee
RedCoast Corporation
05/2024 – 08/2024 | Islamabad, Pakistan

- Analyzed **8+** datasets with Python/SQL/BigQuery; improved pipeline throughput **60%**.
- Consolidated backend logic to speed up dashboard loads **20%**.
- Ran **6** multivariate A/B tests across **4** segments; lifted **CTR +0.3%** in **Q1**.

Intern
PTCL
06/2023 – 12/2023 | Islamabad, Pakistan

- Delivered a task-management app in C#, ASP.NET, SQL for **10 internal users** on time and within scope.
- Introduced role-based workflows; cut task handoff time **5%**.
- Added time-tracking and performance views to improve accountability.

PROJECTS

Apex Analytics
React | Django | GCP | Docker

A powerful analytics tool designed for data customization and visualization.

- Architected and deployed a scalable data analytics platform using a **Django/PostgreSQL** backend and React frontend, integrating over **100 diverse systems** to centralize data aggregation.
- Utilized **BigQuery** and **GCP** services to optimize data pipelines, ensuring fast, scalable performance for dynamic dashboards and real-time KPI monitoring.
- Developed a robust REST API to facilitate advanced customization options and generated custom, exportable reports, enhancing data- driven decision-making processes.
- Implemented Docker containerization for seamless deployment and ensured enterprise-grade data security and compliance standards for sensitive information.

EvaluaSys Al
React | Flask | REST API | MongoDB | Al Model Integration | Al Implementation | Speech-to-Text | Text-to-Speech | NLP

A large-scale web portal providing Al-driven, transparent assignment evaluation via live project demos, quizzes, and plagiarism detection.

- Developed an end-to-end evaluation portal using **React**, **Flask**, and **MongoDB** to automate project assessment, moving beyond traditional manual checking and test-case evaluations.

- Engineered core Al features including real-time Facial Detection and Expression Evaluation during live demos to ensure transparency and prevent malpractice.
- Integrated **Natural Language Processing** (NLP) capabilities (Speech-to- Text & Text-to-Speech) to facilitate Al-based student Q&A verification, handling both voice and text inputs seamlessly.
- Streamlined the evaluation workflow by implementing automated project submission checking and dynamic Al quiz generation tailored to the uploaded project content.

unHinged
Godot Engine | C# | AI Systems | Procedural Generation | State Machines | Object Pooling | Physics Engine | Audio Systems
A psychological 2D survival escape game that uses trauma and memory loss as core gameplay mechanics.
- Architected and developed a 2D psychological survival game using **Godot Engine** and **C#**, successfully delivering the project as a Final Year Project.
- Engineered a dynamic "**Fear Meter**" **system** that actively distorts visual/audio environments and gameplay mechanics based on the protagonist's mental state.
- Implemented procedural generation algorithms for map layouts ("**Memory Maps**") and utilized advanced **Object Pooling** techniques to maintain a consistent **60 FPS** on low-end hardware.
- Developed sophisticated AI systems with behavior trees and state machines to create unpredictable encounters that respond dynamically to player psychology.

Avenix
React | TypeScript | Express.js | Node.js | Python | PostgreSQL | LangChain | Vector Stores | Embeddings | AI/ML | REST APIS
A cutting-edge AI platform delivering comprehensive analytics, advanced LLM integration, and real-time data visualization.
- Architected and deployed a dual-role, enterprise-ready AI platform using a **React/TypeScript** frontend and a **Node.js/Express.js/PostgreSQL** backend, containerized with Docker.
- Integrated **LangChain** framework and **Python** microservices to enable advanced chat capabilities, contextual memory, and intelligent automation via **LLMs**.
- Engineered a high-performance system utilizing **WebSockets** and Redis **pub/sub** for real-time data synchronization, achieving sub-second latency and a **40% improvement** in query load times.
- Delivered the production-ready platform three months ahead of schedule, implementing robust **RESTful APIS** and an intuitive UI for complex data visualization and decision-making.

Raah-e-Raast
Android (Native) | Kotlin | Firebase | Google Cloud API A native Android platform connecting tourists with verified local guides for authentic travel experiences.
- Developed a native Android application using **Kotlin** and the **MVVM architecture**, acquiring a user base of over **1,000 active users** since launch in 2023.
- Integrated **Google Maps API** and **GPS** services to implement real-time location tracking, live location sharing, and robust geofencing for user safety and route optimization.
- Leveraged **Firebase** (Firestore, Realtime DB, Auth) for secure authentication, instant data synchronization, and push notifications within a functional booking system.
- Optimized location tracking services to reduce battery consumption by **40% using intelligent update** intervals and background service optimization.

AdSync Pro
Python | Django Stack | Next.js | MongoDB | LLM Prompt Engineering
A modern ad-management and automation platform designed to streamline campaign creation, tracking, and performance monitoring.
- Engineered a full-stack ad-management platform utilizing a **Django/Python** backend and **Next.js** frontend, enabling streamlined campaign creation and tracking.
- Implemented a scalable architecture with **MongoDB** integration for efficient data handling of real-time campaign analytics and performance metrics.
- Applied **LLM Prompt Engineering** to automate aspects of ad creation or management workflows, enhancing platform efficiency and speed.
- Designed a clean, intuitive UI with secure authentication, providing users with comprehensive dashboards for monitoring and managing ad campaigns.

SnapReceipt AI
Python | Django Stack | MongoDB | LLM Prompt Engineering
A smart receipt-management platform that automates capturing, organizing, and tracking purchases using AI.
- Developed an automated receipt-management system using **Python/Django** and **MongoDB** to streamline expense tracking and financial record organization.
- Integrated AI capabilities via **LLM Prompt Engineering** to instantly extract key data (merchant, date, total) from uploaded receipts with **high accuracy**.
- Designed a secure, searchable dashboard interface allowing users to categorize, tag, and export financial records efficiently, reducing manual data entry.
- Created a robust platform that simplifies expense tracking through AI-driven automation of data capture and organization.

LANGUAGES

- English
- Urdu

INTERESTS

- Travelling
- Book Reading
- Gaming
- Learning New Skills